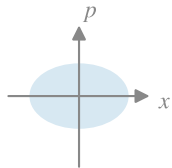


The gauge orbit of the photon dressing



$$\langle a^\dagger a \rangle_\lambda$$

$$U_\lambda = \exp[i\lambda G(a + a^\dagger)]: x \rightarrow x + 2\lambda G, \quad p \text{ unchanged}$$

$$\langle a^\dagger a \rangle_\lambda = \langle a^\dagger a \rangle_0 + C\lambda + Q\lambda^2$$

irreducible

Coulomb

λ_*

Dipole

gauge parameter λ